UNION PUBLIC SERVICE COMMISSION

Civil Services (Preliminary Exam) - 2023

CSAT : PAPER-II

(Analysis with Explanation)

Time : 2 hours **Maximum Number : 200**

Passage-1

We often hear about conflicts among different states n India over river waters. Of the 20 major river systems, 14 are already water-stressed; 75% of the population lives in water-stressed regions, a third of whom live in water-scarce areas. Climate change, the demands of rising population and the need for agriculture to keep pace, and increased rate of urbanization and industrialization will exacerbate water stress. According to the constitution of the India, water s a state subject and not that of the Union, except for regulation of inter-state rivers. Key to ensuring balance between competing demands of various stakeholders is a basin-based approach to allocate water amongst constituent regions and states. Allocating fair share of water to them requires assessments based on objective criteria, such as specificities of the river basin, size of dependent population, existing water use and demand, efficiency of use, projected future use, etc. while ensuring the environmental needs of the river and aquifers.

1. Which one of the following statements best reflects the most rational, practical and immediate actions required to ensure fair and equitable allocation of water to different stakeholders?

(a) A national, pragmatic, legal and policy framework for water allocation should be made.
(b) all river systems of the country should be linked and huge aquifers created.
(c) Water channels between regions of water surplus and regions of water deficit should be created.
(d) To mitigate water crisis, water demand of sectors such as agriculture and industry should be reduced.

Ans. (A) : According to the passage, a national. pragmatic, legal and policy framework for water allocation should be made for the most rational, practical and immediate action to ensure fair and equitable allocation of water to different stakeholders.

Passage-2

More than half of Indian women and almost a quarter of Indian men of working age suffer from anemia. According to studies, they are anywhere from 5-15% less productive than they could, be, as a result thereof. India also has the largest tuberculosis burden in the would, costing 170 million workdays to the country annually. But what is just as important as lost productivity now is lost potential in the future. It is becoming increasingly clear that on many measures of cognitive ability, malnourished India children perform two or three times worse than their adequately nourished peers. For an economy that will be more dependent on highly skilled workers, this poses a significant challenge. And it is one that really should be addressed given India's demographic outlook.

2. Which one of the following statements best reflects what is implied by the passage?

(a) Education system must be strengthened n rural areas.
(b) Large scale and effective implementation of skill development programme is the need of the hour.
(c) For economic development, health an nutrition of only skilled workers needs special attention.
(d) For rapid economic growth as envisaged by us, attention should be paid to health and nutrition of the people.

Ans. (d) : According to the passage, for rapid economic growth as envisaged by us, attention should be paid to health and nutrition of the people.

Passage-3

In India, a majority of farmers are marginal and small, less educated and possess low adaptive capabilities to climate change, perhaps because of credit and other constraints. So, one cannot expect autonomous adaptation to climate change. Even if it was possible, it would not be sufficient to offset losses from climate change. To deal with this, adaptation to climate change is paramount, alongside a fast mitigation response. Another solution is to have a planned or policy-driven adaption, which would require the government to come up with policy recommendations. Perception is a necessary pre-requisite for adaptation. Whether farmers are adapting agricultural practices to climate change depends on whether they perceive it or not. However, this is not always enough for adaptation. It is important how a farmer perceives the risks associated with climate change.

3. Which one of the following statements best reflects the most logical and rational message conveyed by the author of the passage?

(a) Adaptation to climate change and mitigation response are basically the responsibilities of the government.
(b) Climate change causes a change in government policies regarding land use patterns in the country.
(c) Risk perceptions of farmers are important for motivating them for taking adaptation decisions.
(d) Since mitigation is not possible, governments should come up with policies for quick response to climate change.

Ans. (c) : According to the passage, risk perceptions of farmers are important for motivating them for taking adaptation decisions; is the most logical and rational message conveyed by the author.

4. Raj has ten pairs of red, nine pairs of white and eight pairs of black shoes in a box. If he randomly picks shoes one by one (without replacement) from the box to get a red pair of shoes to wear, what is the maximum number of attempts he has to make?

(a) 27 (b) 36
(c) 44 (d) 45

Ans. (d) : According to the given,
Pairs of red shoes = 10
Pairs of white shoes = 9
Pairs of black shoes = 8
Let Raj drowns 10 red shoes (all of left foot), 18 white shoes and 16 black shoes.
Total number of shoes drawn = 44
But still red pair is not complete
Therefore, one more shoe drawn from the box will complete the red pair.
44 + 1 = 45

5. In how many ways can a batsman score exactly 25 runs by scoring single runs, fours and sixes only, irrespective of the sequence of scoring shots?

(a) 18 (b) 19
(c) 20 (d) 21

Ans. (b) : According to the question, 25 runs is to be made with single runs, fours and sixes only. Therefore,

	1s	4s	6s
1	25	0	0
2	21	1	0
3	17	2	0
4	13	3	0
5	9	4	0
6	5	5	0
7	1	6	0
8	19	0	1
9	15	1	1
10	11	2	1
11	7	3	1
12	3	4	1
13	13	0	2
14	9	1	2
15	5	2	2
16	1	3	2
17	4	0	3
18	1	1	3
19	1	0	4

Hence, 19 ways.

6. There are four letters and four envelopes and exactly one letter is to be put in exactly one envelope with the correct address. If the letters are randomly inserted into the envelopes, then consider the following statements:

1. It is possible that exactly one letter goes into an incorrect envelope.
2. There are only six ways in which only two letters can go into the correct envelopes.

Which of the statements given above is/are correct?

(a) 1 only (b) 2 only
(c) Both 1 and 2 (d) Neither 1 nor 2

Ans. (b) : **Statement 1** - It is not possible that exactly one letter goes into on incorrect envelope. If 3 letters go in correct envelope then 4[th] will also go in correct envelope.
Statement 2 - Number of ways in which only two letters can go into the correct envelopes out of 4 letter

$$= {}^4C_2 = \frac{4!}{2!\times 2!} = \frac{4\times 3}{2} = 6 \text{ ways}$$

7. What is the remainder when

(a) 0 (b) 1
(c) 2 (d) 4

Ans. (a) : According to the question,

$$= \frac{85\times 87\times 89\times 91\times 95\times 96}{100}$$

$$= \frac{-15\times -13\times -11\times -9\times -5\times -4}{100} = \frac{386100}{100}$$

Remainder = 0

8. What is the unit digit in the expansion of $(57242)^{9\times 7\times 5\times 3\times 1}$

(a) 2 (b) 4
(c) 6 (d) 8

Ans. (a) : According to the question,
$(57242)^{9\times 7\times 5\times 3\times 1}$
$(57242)^{945}$
By dividing the power 945 by 4 we get Remainder = 1
Therefore, $(57242)^1 = 57242$
Hence, unit digit = 2

9. If ABC and DEF are both 3-digit numbers such that A, B, C, D, E and F are distinct non-zero digits such that ABC + DEF = 1111, then what is the value of A + B + C + D + E + F + ?

(a) 28 (b) 29
(c) 30 (d) 31

Ans. (d) : According to the question,

$$\begin{array}{cccc} & A & B & C \\ + & D & E & F \\ \hline 1 & 1 & 1 & 1 \end{array}$$

C + F must be equal to 11
B + E must be equal to 10 (because 1 is carryover)
A + D must be equal to 10
Therefore, A + D + B + E + C + F
=10 + 10 +11
= 31

10. D is a 3-digit number such that the ratio of the number to the sum of its digits is least. What is the difference between the digit at the hundred's place and the digit at the unit's place of D?

(a) 0 (b) 7
(c) 8 (d) 9

Ans. (c) : For the ratio of the number to the sum of its digits to be least, the value of D = 199

Then, ratio $= \frac{199}{19} = 10\frac{9}{19}$

Therefore, difference between the digit at hundred's place and the digit at units place of D = 9–1 = 8